Mise

Test a restaurant idea on a simulated Pittsburgh before you open it.

Team Peanuts

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Track

DoorDash, Back House

Event

NOVA 2026, ScottyLabs

3 to 5% margins

A new menu, a price change, later hours, a lower delivery fee. Owners make these calls with records of what already happened, and no way to try one first.

Margin figure from DoorDash's NOVA problem statement

Who it's for

Two owners we designed for

Persona · opening a place

Late-night chicken near Pitt

Has a concept and a lease to look at on Forbes. Doesn't know if students will actually order at midnight, or what to charge.

"Is Friday night on Forbes enough, and who am I taking customers from?"

Persona · already open

Noodle shop in Shadyside

Thinking about cutting the delivery fee and staying open later. Every test costs a real week of revenue.

"Who does a $1.99 fee actually bring in, and is it worth it?"

01

Describe the idea in chat, or import a real restaurant

02

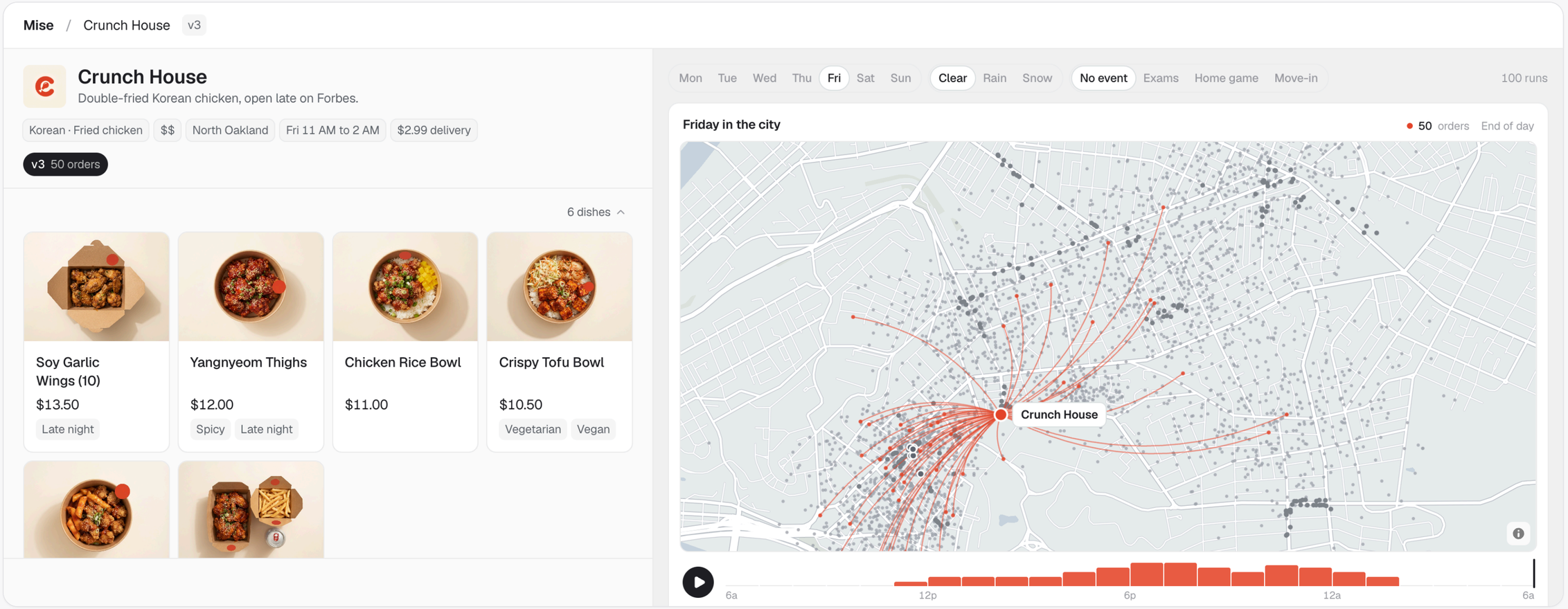
Mise drafts it: menu, prices, logo, dish photos

03

Simulate a day in the city, hour by hour

04

Set a goal and approve the edits that get there



Live demo

Happy path

1. Describe a late-night Korean chicken spot near Pitt
2. Mise drafts the menu, logo and photos
3. Run Friday: orders, revenue, who said no
4. "Get me to 90 orders on Friday"

Sad path

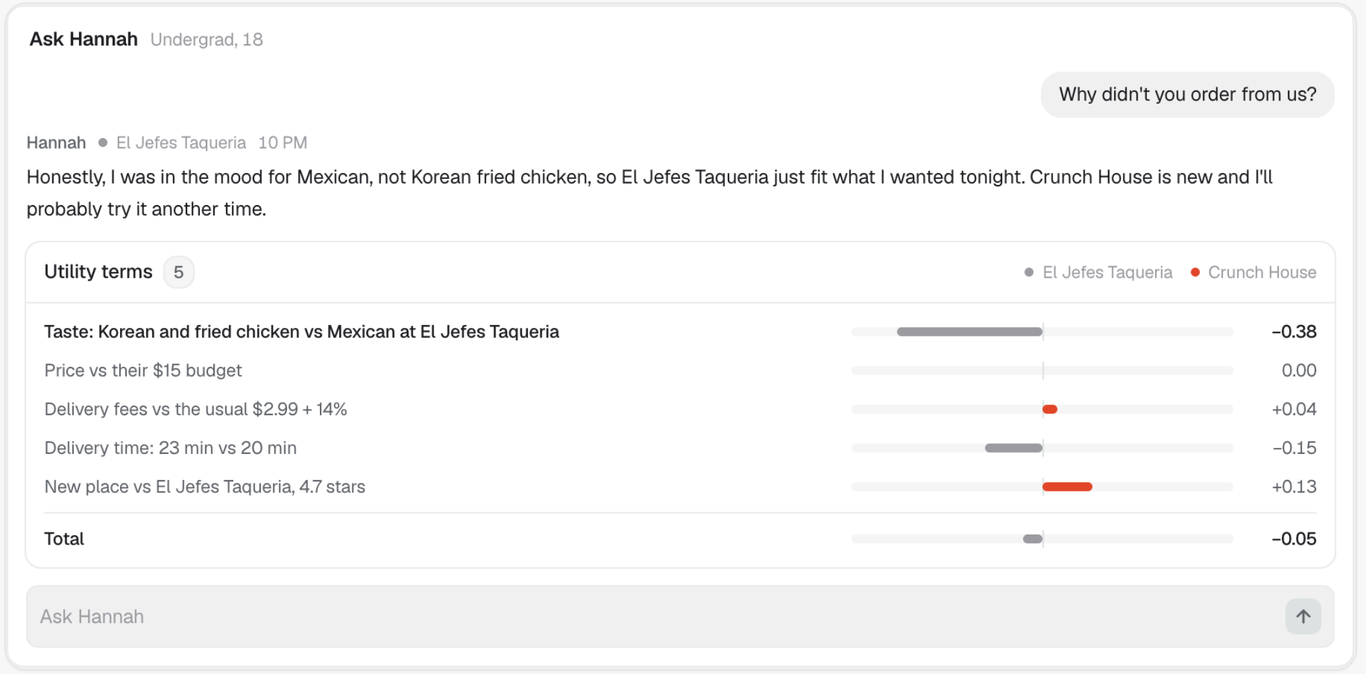
1. The idea underperforms on Friday
2. Mise shows who passed and why
3. Ask a resident directly
4. Import a real place and check the sim against Google

A city of 10,000 people

Every hour, each simulated resident decides whether to order, cook or skip, and where from, out of **192 real Pittsburgh restaurants** plus yours.

The choice comes from taste, price against their budget, delivery fee, delivery time and ratings. Change the day, the weather, or a campus event and run it again.

You can ask any resident why. Their answer is built from the same terms the simulator used.



Accuracy

How we know the numbers hold up

Matches real restaurants	Fit to Google busy times for 51 places, tested on 13 it never saw	r 0.54 vs 0.44 baseline
Behaves like a city	Automatic checks for direction, double counting and consistency	26 of 26 pass
Stable headline	100 simulated Fridays per prediction instead of 20	spread 12 → 6 orders
Fair before and after	Every version faces the same residents and the same random draws	only the edit differs
Optimizer re-checked	Final edits re-scored on simulated days the search never saw	beat original 10 of 10
Model proposes, code decides	Edits are clamped by code, scored by the simulator, approved by the owner	predictions never from the LLM

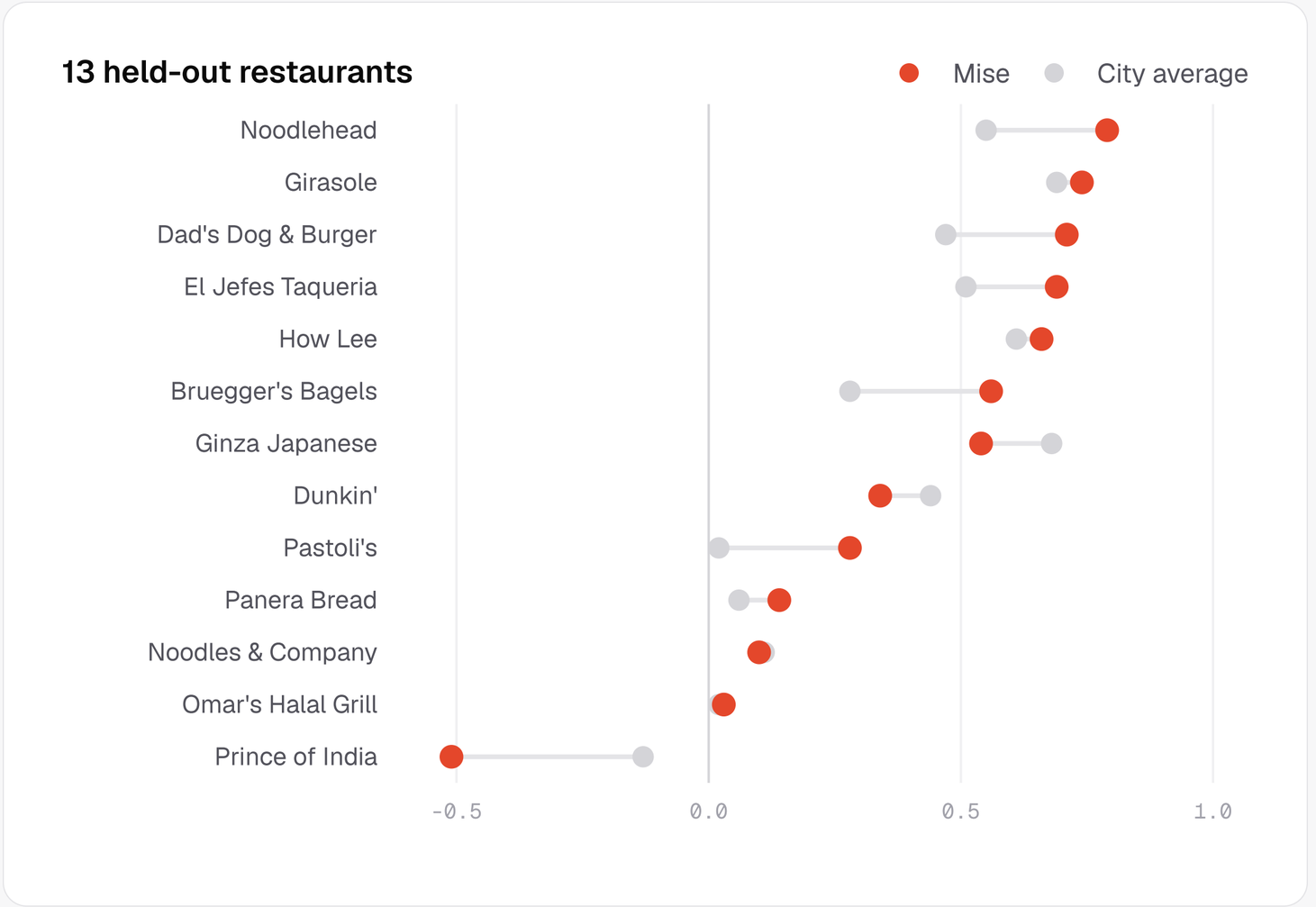
It matches real restaurants it never saw

0.54 Mise, calibrated

0.44 City-average curve for every place

0.32 Mise before calibration

Median correlation with each place's real hourly curve. We fit on 51 restaurants and tested on 13 the model never saw.



26 checks run on every change

Direction

Cheaper means more orders. Pricier, farther, or closing early means fewer. Free delivery and a launch promo mean more. Monday is below Friday. A vegan main cuts diet objections.

Words can't move numbers

Editing only the tagline gives identical orders.

One story

For sampled residents, the list, the map and the interview agree on what they ordered.

Import a real restaurant unchanged

It should get back its own orders and take nothing from itself.

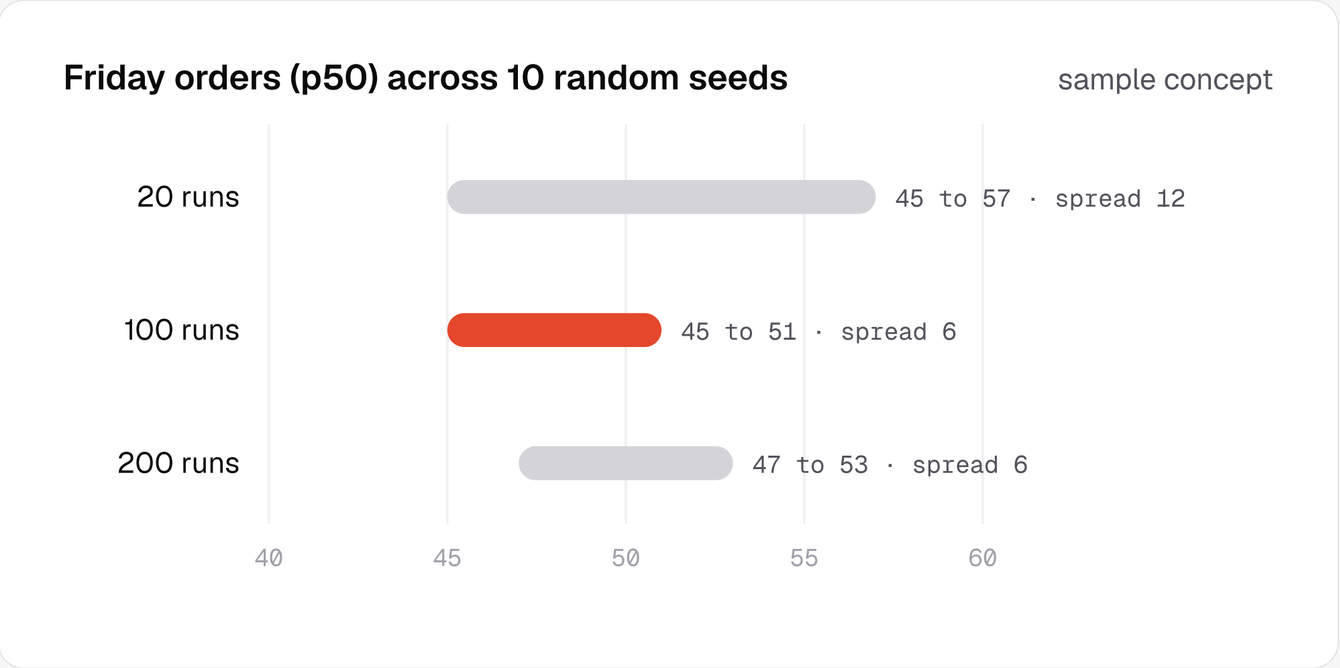
	Today in the sim	Imported as-is	Taken from itself
Noodlehead	46	45	0
CHiKN	66	69	0
Pizza Palermo	53	55	0
Sushi Fuku	9	10	0

Enough runs that the headline doesn't wobble

Each prediction simulates many Fridays. Every run draws a city-wide demand level, how fast people find a new place, and every resident's orders.

At 20 runs, the headline moved by up to **12 orders** between random seeds. At 100 runs it moves by **6**, and 200 runs doesn't improve on that.

So every prediction is now 100 runs, about 1.4 s.

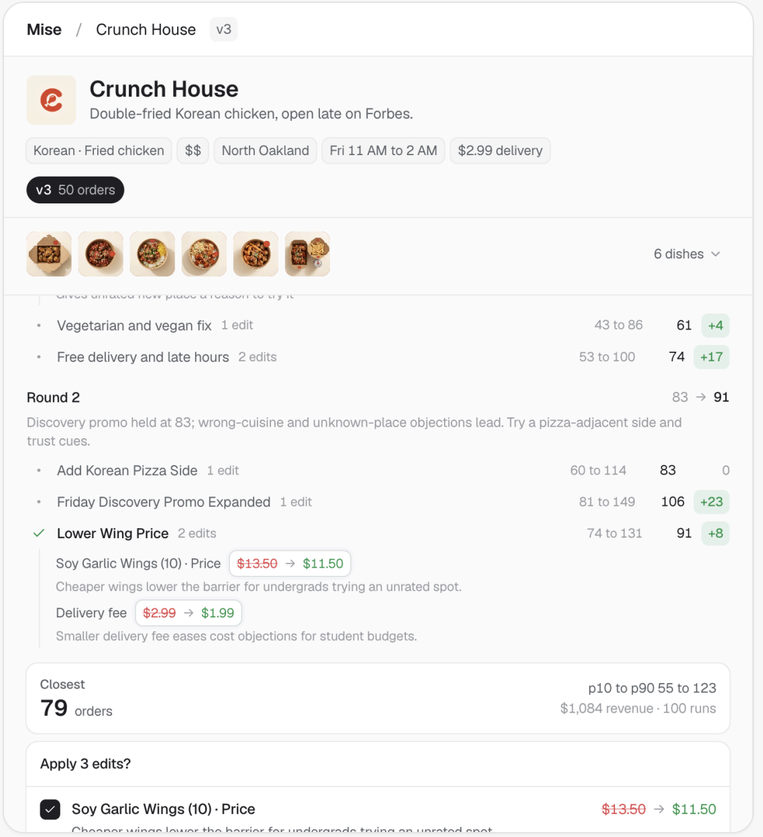


The optimizer is checked on days it never saw

The search tries edit sets and keeps the best score. The best of several noisy scores runs high. So the number the owner sees comes from a fresh check: 100 new Fridays, original and edited concept on the same days.

Search's own score	91 orders
Fresh check, shown to the owner	79 orders
10 more fresh seeds, beat the original	10 of 10
Median across those seeds	49.5 → 83

Same goal, 90 orders on Friday. The 10-seed check is a scripted rerun (`scripts/verify`). Code limits every edit: 3 per set, no dish under 60% of the owner's price, discounts up to 30%, fee up to $5.99. Nothing applies until the owner approves.



Trade-offs

What's real and what's estimated

Real

192 restaurants from OpenStreetMap and Google Maps

Google popular times for 64 of them, collected today

Menus pulled from the restaurant's own site when every item and price is on the page

Estimated

The resident mix per neighborhood is hand-set. Census data is next

Google shows how busy a place is, not order counts, so we match the shape of the day, not volume

Price and fee sensitivity are hand-set from typical delivery-app behavior, not fitted

Menus without a website are drafted and labeled that way

Stack TypeScript, Bun + Hono, React, MapLibre + deck.gl. Claude Sonnet for chat, Claude Haiku for the goal loop and interviews, Gemini for images, through OpenRouter.

With DoorDash

What this becomes with real order data

Calibrate on real orders	Swap Google's busy curves for each merchant's order history, so the model learns volume and price sensitivity, not only timing.
More cities	Pittsburgh is the first. Another city needs its restaurant list, busy times and resident mix.
Test before you change it	An owner tries a price, hours or fee change in the merchant tools and sees the result before turning it on.

Thanks!

github.com/ZhYGuoL/mise